Euclid's Elements — Book II
Proposition 12

Formal Reconstruction — Technical Documentation

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In obtuse-angled triangles the square on the side subtending the obtuse angle is greater than the squares on the sides containing the obtuse angle by twice the rectangle contained by one of the sides about the obtuse angle, namely that on which the perpendicular falls, and the straight line cut off outside by the perpendicular towards the obtuse angle.

1 Introduction

Proposition II.12 is the first proposition of Book II in which Euclid starts from a triangle with an *obtuse* angle and compares the square on the opposite side with the squares on the two sides containing that angle. In the lettering used by the current formal theorem, the triangle is ABC, the obtuse angle is BAC, and the perpendicular from B meets the carrier of AC at a point D lying beyond A:

$$D - A - C, \qquad BD \perp AC.$$

Euclid's content statement is

$$\mathrm{Sq}(BC) = \mathrm{Sq}(BA) + \mathrm{Sq}(AC) + 2\ \mathrm{Rect}(CA, AD).$$

Equivalently, in Euclid's wording, the square on BC is *greater by* twice the rectangle contained by CA, AD than the two squares on BA, AC.

The current Lean theorem reconstructs this statement without numerical area, segment multiplication, coordinates, trigonometry, or a numerical angle measure. It constructs concrete square representatives on AB, AC, and CB, a concrete rectangle representative contained by DA, AC, and proves the exact additive relation in the `HilbertScissorsEquicomplementable` layer.

A modern algebraic shadow is the obtuse case of the law of cosines. That interpretation is useful only as a mnemonic check. It is not the proof architecture of either Euclid or the Lean reconstruction. The actual source DAG is:

$$\text{I.12} \longrightarrow \text{II.4} \longrightarrow \text{Common Notion 2} \longrightarrow \text{I.47 twice}.$$

The production source begins with

```
import CGJteamLab.HilbertRightAngle
import CGJteamLab.Proposition12
import CGJteamLab.Proposition16
import CGJteamLab.Proposition47
import CGJteamLab.HilbertSquareTransport
import CGJteamLab.Proposition2_4
```

Thus Proposition II.12 has one direct earlier Book II theorem dependency, namely II.4. The essential Book I dependencies are I.12, I.16, I.46, and I.47, together with the established Hilbert incidence, order, congruence, parallel, and scissors infrastructure.

2 Euclid's Statement and Classical Configuration

2.1 The statement

In Heath's wording, Proposition II.12 states:

> In obtuse-angled triangles the square on the side subtending the obtuse angle is greater than the squares on the sides containing the obtuse angle by twice the rectangle contained by one of the sides about the obtuse angle, namely that on which the perpendicular falls, and the straight line cut off outside by the perpendicular towards the obtuse angle.

For the triangle ABC with obtuse angle BAC, produce the side CA beyond A and let BD be perpendicular to the produced carrier. The source order is

$$D - A - C.$$

The final content identity is

$$\mathrm{Sq}(CB) = \mathrm{Sq}(AB) + \mathrm{Sq}(AC) + 2\ \mathrm{Rect}(CA, AD).$$

The orientation of the final rectangle is not mathematically essential, but it is formally visible. The production theorem chooses a representative contained by DA, AC; the side reversal and rectangle-representative independence are handled inside the established content transport layer.

2.2 The classical construction

The construction is short.

(1) Start with the nondegenerate triangle ABC and assume that $\angle BAC$ is obtuse.

(2) Produce CA through A.

(3) From B draw BD perpendicular to the produced line CA by I.12.

(4) The resulting order is
$$D - A - C.$$

The rest of the proposition is a content argument. The squares and rectangle appearing in the proof are not additional geometric hypotheses; they are the figures "on" or "contained by" the corresponding segments in the standard Book II sense.

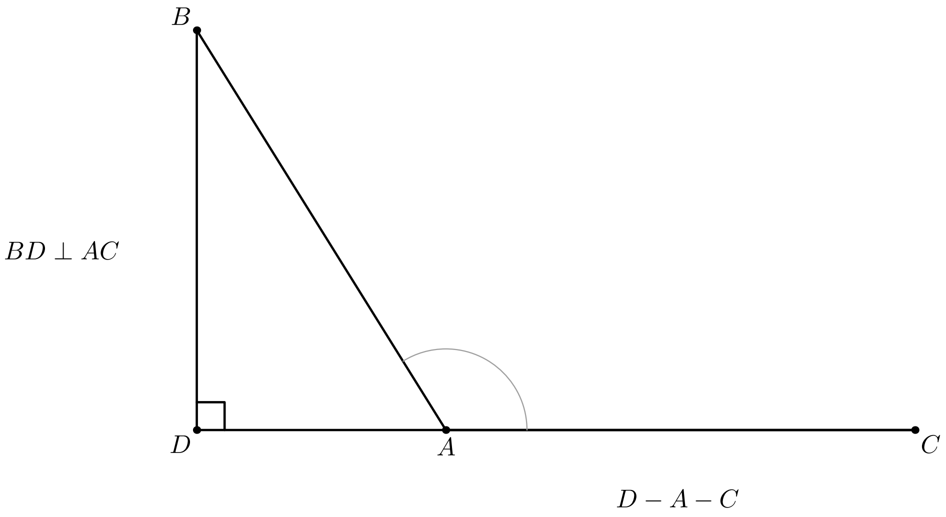

Figure 1: The classical starting configuration for Proposition II.12 in the lettering of the Lean source. The angle BAC is obtuse, BD is perpendicular to the carrier AC, and the source order is $D - A - C$. The light arc records only the qualitative obtuse-angle hypothesis. No square or rectangle is shown yet: this figure states the geometric input rather than the later content calculation.

Diagrammatic audit. The starting configuration is not sufficient to explain the proof. Three further figures are retained below because the geometric state genuinely changes: the reductio branch used to prove the position of the perpendicular foot, the pair of right triangles sharing the DB square, and the II.4 square dissection on DC cut at A. No separate figure is introduced merely for representation transport or for the final associative-commutative regrouping.

3 Classical Proof

Euclid's proof is a direct comparison of one II.4 decomposition with two Pythagorean decompositions.

Since CD is cut at A, Proposition II.4 gives

$$\mathrm{Sq}(DC) = \mathrm{Sq}(DA) + \mathrm{Sq}(AC) + 2\ \mathrm{Rect}(DA, AC).$$

The rectangle may equally be read as contained by CA, AD; the present orientation anticipates the Lean representative.

Add the square on DB to both sides. This is Euclid's use of Common Notion 2:

$$\mathrm{Sq}(DC) + \mathrm{Sq}(DB) = \mathrm{Sq}(DA) + \mathrm{Sq}(DB) + \mathrm{Sq}(AC) + 2\ \mathrm{Rect}(DA, AC).$$

The triangle DCB is right at D, so I.47 gives

$$\mathrm{Sq}(CB) = \mathrm{Sq}(DC) + \mathrm{Sq}(DB).$$

The triangle DAB is also right at D, so I.47 gives

$$\mathrm{Sq}(AB) = \mathrm{Sq}(DA) + \mathrm{Sq}(DB).$$

Substituting these two equalities into the previous identity yields

$$\mathrm{Sq}(CB) = \mathrm{Sq}(AB) + \mathrm{Sq}(AC) + 2\ \mathrm{Rect}(DA, AC),$$

which is the required proposition.

The local classical dependency picture is therefore

$$\begin{gathered}
\text{triangle } ABC,\ \angle BAC \text{ obtuse} \\
\Downarrow \\
\text{I.12: construct } BD \perp CA \text{ produced} \\
\Downarrow \\
D - A - C \\
\Downarrow \\
\text{II.4 on } D - A - C \\
\mathrm{Sq}(DC) = \mathrm{Sq}(DA) + \mathrm{Sq}(AC) + 2R \\
\Downarrow \\
\text{Common Notion 2: add } \mathrm{Sq}(DB) \\
\Downarrow \\
\text{I.47 on } DCB \quad + \quad \text{I.47 on } DAB \\
\Downarrow \\
\mathrm{Sq}(CB) = \mathrm{Sq}(AB) + \mathrm{Sq}(AC) + 2R \\
\Downarrow \\
\text{II.12.}
\end{gathered}$$

Two features are worth emphasizing. First, there is no cancellation step in II.12. The entire content calculation is forward composition and addition. Second, Euclid's short proof presupposes the positional fact $D - A - C$. In the formal reconstruction that fact is a theorem, not a visual convention.

4 The Geometric Identity in the Current Formalization

The public theorem takes

```
(A B C : Geo.Point)
(hABC : Not (PrimCollinear Geo A B C))
(hObtuse : HilbertObtuseAngle Geo B A C)
```

and constructs all other geometric and representation data internally.

At the final interface Lean produces a point D satisfying

$$D - A - C,$$

the two right angles

$$\angle ADB, \qquad \angle CDB,$$

concrete square representatives on

$$AB, \qquad AC, \qquad CB,$$

and a concrete rectangle representative R contained by DA, AC.

The conclusion is

$$S_{CB} \simeq_{\mathrm{ec}} (S_{AB} + S_{AC}) + (R + R),$$

where $\simeq_{\mathrm{ec}}$ denotes `HilbertScissorsEquicomplementable`.

This is not an inequality in an ordered area field. Euclid's phrase "greater by twice the rectangle" is represented by an exact additive equal-content statement. The theorem therefore records not merely that one content is larger than another, but an explicit positive decomposition of the larger content into the smaller content plus two copies of the rectangle.

The conclusion is also not strengthened to `HilbertScissorsEq`. The existing I.47 interface returns equicomplementability, and the final theorem remains at that natural content level. Exact scissors equalities occur internally inside the II.4 block.

5 Reusable Book II Infrastructure Used

5.1 Proposition II.4

The sole direct earlier Book II theorem is `euclid_proposition_2_4`. Its production API is configured: it expects a square, two rectangle representatives, two auxiliary II.3 rectangles, their cut data, squares on the two parts, and the two cross rectangles.

The local wrapper

```
proposition2_12_ii4_DAC
```

specializes that API to the order

$$D - A - C.$$

Under the substitution

$$(A, B, C)_{\mathrm{II.4}} = (D, C, A),$$

the II.4 conclusion becomes

$$S_{DC} \simeq_{\mathrm{ec}} (S_{AD} + S_{AC}) + (R + R).$$

The two cross-rectangle arguments of II.4 represent the same "twice the rectangle" term. The wrapper therefore supplies the same concrete representative twice and uses only associative–commutative regrouping to place the result in the II.12 order.

5.2 Existential packaging of the II.4 diagram

The higher-level helper

```
proposition2_12_ii4_exists
```

starts only from

$$D - A - C$$

and constructs the complete geometric data required by the configured II.4 API. It constructs:

- a square on DC;
- the internal cut of that square at A;
- arbitrary representatives of the rectangles contained by DC, DA and DC, AC;
- the two auxiliary rectangles used by the II.3 blocks inside II.4;
- their internal cut diagrams;
- squares on AD and AC;
- one representative of the rectangle contained by DA, AC.

All these witnesses are hidden from the public theorem. The wrapper returns only the square representatives and the final II.4 content relation needed by the subsequent Pythagorean substitution.

5.3 Square representative transport

I.47, I.46, and the II.4 construction create independent concrete squares. The formal proof must therefore identify their contents explicitly.

Three representative problems occur:

(1) the two I.47 calls construct two different squares on DB;

(2) I.47 and II.4 construct different squares on DC;

(3) I.47 naturally uses a square on DA, while II.4 returns a square on the oppositely oriented segment AD.

The theorem `hilbert_square_transport` resolves all three situations. No independently constructed witnesses are identified by definitional equality.

5.4 Equicomplementability addition and reassembly

The helper imported from I.47,

```
i47_aux_equicomplementable_add
```

lifts equal-content relations through formal addition. Together with `equicomplementable_symm`, `equicomplementable_trans`, and reflexivity, it implements the Common-Notion-2 part of the proof.

The proposition-local helper

```
proposition2_12_scissors_final_bridge
```

contains no new geometry. It is the pure content form of Euclid's final substitution.

6 New Proposition-Local Parallelogram-Cut Infrastructure

A major formal feature of II.12 is hidden behind the short public theorem. The low-level II.4 API still requires complete geometric cut diagrams. Proposition II.12 therefore develops a general internal cut constructor for an arbitrary parallelogram.

The proposition-local chain is

```
proposition2_12_parallelogram_L_on_DE
proposition2_12_left_cut_parallelogram
proposition2_12_upper_cut_between
proposition2_12_diagonal_cut_intersection
proposition2_12_parallelogram_cut_exists
```

Start with a parallelogram $BCED$ and an interior point M:

$$B - M - C.$$

The target is to construct points L, X such that

$$D - L - E, \qquad D - X - C, \qquad M - X - L,$$

and the two cut pieces are parallelograms

$$LDBM, \qquad CELM.$$

The first helper proves that the upper cut point lies on the opposite carrier. Completing CEM to the parallelogram $CELM$ gives

$$EL \parallel MC.$$

The original parallelogram gives

$$ED \parallel CB.$$

Because B, M, C are collinear, both EL and ED are parallel to the same base carrier through E. Euclidean parallel uniqueness forces their carriers to coincide, yielding

$$D, L, E \text{ collinear}.$$

The next helper constructs the complementary parallelogram $LDBM$. The delicate point is $L \neq D$. From $B - M - C$ one has the strict segment comparison

$$CM < CB.$$

Opposite-side congruences in the two parallelograms transport this to

$$EL < ED,$$

hence $L \neq D$. The required two pairs of opposite parallel sides then follow by collinearity and parallel transport.

The order helper strengthens collinearity to

$$D - L - E.$$

The three relevant corresponding segment congruences are

$$BM \cong DL, \qquad BC \cong DE, \qquad MC \cong LE.$$

Hilbert's three-point order transport sends the known order $B - M - C$ to the opposite side.

Finally

```
proposition2_12_diagonal_cut_intersection
```

uses Pasch in triangle CBD. The cut line ML enters through CB at M and cannot meet BD because

$$DB \parallel ML.$$

It must therefore meet CD at an interior point X. Parallel crossing order then gives

$$D - X - C, \qquad M - X - L.$$

The wrapper

```
proposition2_12_parallelogram_cut_exists
```

packages exactly the witnesses required by the existing Book II configured cut APIs.

This theorem is structurally more general than II.12 because it applies to an arbitrary parallelogram, not merely a square or rectangle. It remains in `Proposition2_12.lean` under a proposition-specific name. Promoting it to a permanent general rectangle/parallelogram API should be a separate architectural decision rather than an automatic consequence of its use here.

7 Proof Architecture of the Reconstruction

7.1 Step 1: encode the obtuse angle synthetically

The production module begins with

```
def HilbertObtuseAngle
    [HilbertIncidence Geo]
    (A O B : Geo.Point) : Prop :=
  Exists fun X : Geo.Point =>
    And
      (HilbertRightAngle Geo A O X)
      (HilbertAngleLess Geo A O X A O B)
```

Thus AOB is obtuse when a right angle with the same first arm OA is strictly smaller than AOB. The definition uses only the established synthetic angle-order relation. There is no numerical 90° measure.

The preliminary theorem

```
proposition2_12_obtuse_not_right
```

shows that an obtuse angle cannot itself be right. If both were right, all right angles would be congruent; transport of the strict inequality would produce an angle strictly smaller than itself.

7.2 Step 2: construct the perpendicular foot

The helper

```
proposition2_12_perpendicular_foot_exists_full
```

applies Euclid I.12 to the carrier AC and the external point B. Because ABC is noncollinear, B is off that carrier. The theorem returns points D, R with the incidence data

$$A, D, C \text{ collinear}, \qquad R, D, A \text{ collinear}, \qquad R, A, C \text{ collinear},$$

and a right angle

$$\angle RDB.$$

The auxiliary point R belongs to the interface of I.12. It witnesses the carrier arm of the constructed right angle but does not occur in the final II.12 statement.

7.3 Step 3: prove the source-critical order $D - A - C$

The central geometric theorem is

```
proposition2_12_obtuse_perpendicular_foot_between
```

First Lean rules out $D = A$. If the foot coincided with A, right-angle transport would show that BAC is right, contradicting `proposition2_12_obtuse_not_right`.

The proof then excludes the possibility that D and C lie on the same ray from A. Suppose

$$\mathrm{SameRay}(A, D, C).$$

Extend AD beyond D to a point E. In triangle BAD, I.16 gives an exterior-angle inequality

$$\angle BAD < \angle BDE.$$

Because D and C are on the same ray from A,

$$\angle BAD = \angle BAC.$$

Because ADB is right and $A - D - E$, the exterior angle BDE is right. The obtuse hypothesis also supplies a right angle BAX with

$$\angle BAX < \angle BAC.$$

All right angles are congruent, so

$$\angle BDE \cong \angle BAX.$$

Transporting the first strict inequality therefore yields

$$\angle BAC < \angle BAX.$$

Combining this with

$$\angle BAX < \angle BAC$$

gives a strict angle-order cycle, hence an angle strictly smaller than itself, contradicting irreflexivity.

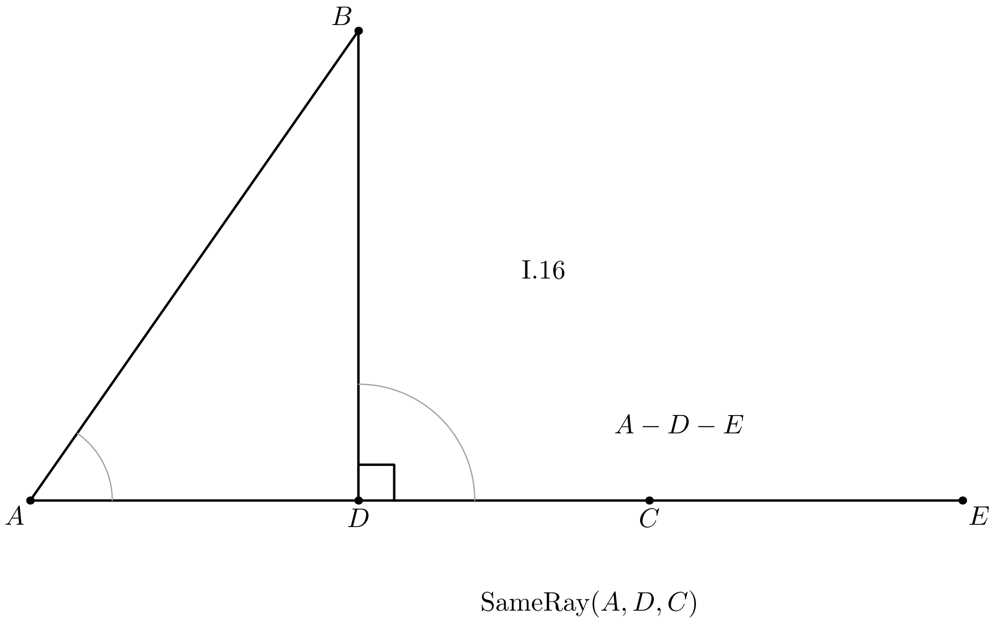

Figure 2: The wrong-position branch excluded by the Lean proof. Assuming that D lies on the same ray from A as C, extend AD through D to E. Triangle BAD then has the exterior angle BDE, which is right, and I.16 gives the strict comparison between the interior angle BAD and that exterior angle. Because AD and AC are the same ray, BAD is the angle BAC relevant to the obtuse hypothesis. The separate obtuse-witness ray AX is deliberately omitted: the displayed branch is a reductio configuration, so no Euclidean picture can realize all contradictory assumptions simultaneously.

Betweenness trichotomy on the collinear triple A, D, C now leaves exactly one possible order:

$$\boxed{D - A - C}.$$

This is the principal proof obligation that is implicit in Euclid's printed diagram but explicit in the formal reconstruction.

7.4 Step 4: obtain both right triangles

The I.12 construction gives a right angle at D with one carrier arm. The helper

```
proposition2_12_right_angle_collinear_transport
```

transports it to

$$\angle ADB.$$

Since $D - A - C$, the rays DA and DC coincide. Same-ray angle transport therefore gives

$$\angle ADB \cong \angle CDB,$$

and right-angle transport yields

$$\angle CDB$$

right as well.

The formal proof also derives the noncollinearity of the two triangles DAB and DCB from the original noncollinearity of ABC and the strict order $D - A - C$.

7.5 Step 5: apply I.47 to triangle DAB

Euclid I.47 gives concrete squares and the equal-content identity

$$S_{AB} \simeq_{\mathrm{ec}} S_{DA} + S^{(1)}_{DB}.$$

The first square on DB is only one representative. Nothing in the type system identifies it with a later square independently erected on the same segment.

7.6 Step 6: apply I.47 to triangle DCB

The second I.47 call gives

$$S_{CB} \simeq_{\mathrm{ec}} S_{DC} + S^{(2)}_{DB}.$$

The helper

```
proposition2_12_normalize_second_pythagoras_DB
```

uses square transport to replace $S^{(2)}_{DB}$ by the representative $S^{(1)}_{DB}$ from the first Pythagorean block.

The wrapper

```
proposition2_12_pythagorean_package_common_DB
```

therefore returns the normalized pair

$$S_{AB} \simeq_{\mathrm{ec}} S_{DA} + S_{DB},$$

$$S_{CB} \simeq_{\text{ec}} S_{DC} + S_{DB}.$$

This is exactly the pair of I.47 identities used in Euclid's classical substitution, now synchronized at the representation level.

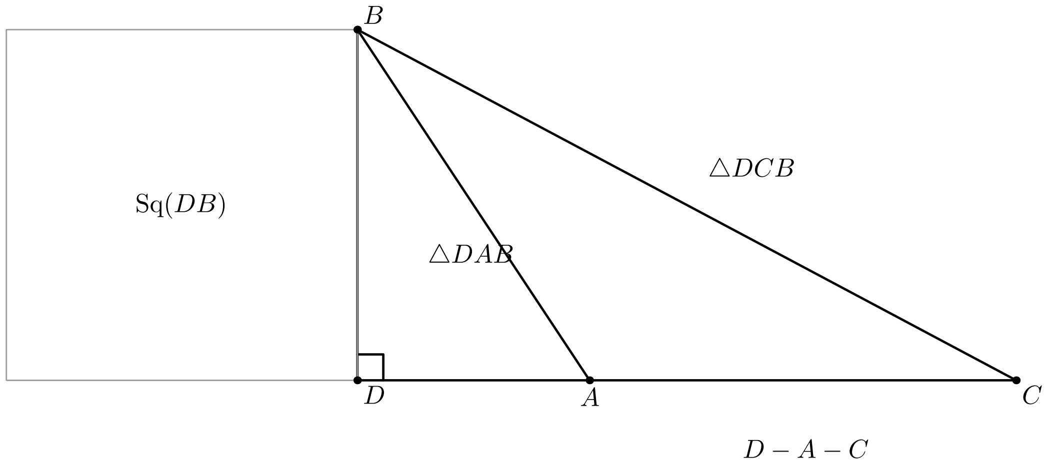

Figure 3: The two right triangles to which I.47 is applied. The strict order $D - A - C$ places DAB inside DCB, and both triangles use the same perpendicular leg DB. The single square drawn on DB makes visible the common summand that must be normalized between the two independent I.47 calls. The other Pythagorean squares are omitted because drawing them would obscure, rather than clarify, the shared-DB geometry.

7.7 Step 7: construct and apply the II.4 block

From the already proved order

$$D - A - C,$$

the theorem

```
proposition2_12_ii4_exists
```

constructs the complete configured II.4 diagram and returns

$$S_{DC}^{(2)} \simeq_{\text{ec}} (S_{AD}^{(2)} + S_{AC}) + (R + R).$$

Here $S_{DC}^{(2)}$ and $S_{AD}^{(2)}$ are independent representatives constructed inside the II.4 block.

7.8 Step 8: normalize the II.4 squares

The theorem

```
proposition2_12_normalize_ii4_squares
```

performs two transports.

First,

$$S_{DC} \simeq_{\text{ec}} S_{DC}^{(2)}$$

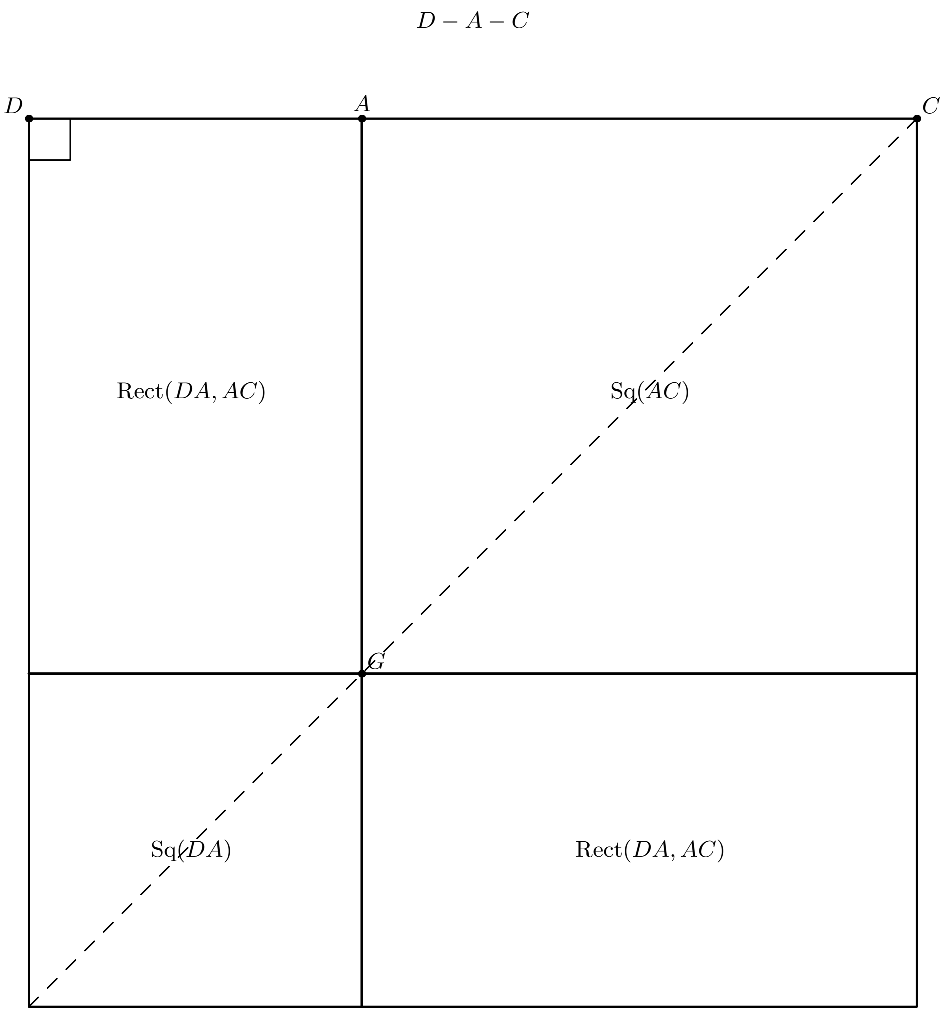

Figure 4: Proposition II.4 applied to the segment DC cut at A. The outer square is the square on DC; the vertical through A and the horizontal through its intersection with the diagonal divide it into the square on DA, the square on AC, and two rectangles contained by DA, AC. The dashed diagonal records the classical II.4 construction. Only the cut point A and the diagonal intersection G are named, because the additional configured cut witnesses constructed by Lean certify this same dissection but do not define new content regions.

because both are squares on the same oriented segment DC.

Second,

$$S_{DA} \simeq_{\text{ec}} S^{(2)}_{AD}.$$

This time the endpoint order is reversed, so the proof explicitly constructs the congruence

$$DA \cong AD$$

before applying square transport.

The normalized II.4 statement is therefore

$$S_{DC} \simeq_{\text{ec}} (S_{DA} + S_{AC}) + (R + R).$$

7.9 Step 9: perform Euclid's final Common-Notion-2 substitution

The pure content helper

```
proposition2_12_scissors_final_bridge
```

takes exactly the three normalized input relations

$$S_{AB} \simeq_{\text{ec}} S_{DA} + S_{DB},$$

$$S_{CB} \simeq_{\text{ec}} S_{DC} + S_{DB},$$

$$S_{DC} \simeq_{\text{ec}} (S_{DA} + S_{AC}) + (R + R).$$

It first adds the unchanged DB square to the II.4 relation:

$$S_{DC} + S_{DB} \simeq_{\text{ec}} ((S_{DA} + S_{AC}) + (R + R)) + S_{DB}.$$

Composing with the second I.47 block gives the same right-hand side for S_{CB}.

The next step is only a formal reassociation and commutation of the finite multiset sum:

$$((S_{DA} + S_{AC}) + (R + R)) + S_{DB} = (S_{DA} + S_{DB}) + (S_{AC} + (R + R)).$$

Lean proves this with

```
ac_rfl
```

which is purely syntactic bookkeeping. It contributes no geometric or content theorem.

Finally the first I.47 block, used symmetrically, replaces

$$S_{DA} + S_{DB}$$

by S_{AB}. Transitivity gives

$$S_{CB} \simeq_{\text{ec}} (S_{AB} + S_{AC}) + (R + R).$$

No cancellation theorem is used anywhere in this final bridge.

Diagrammatic audit. Steps 8 and 9 introduce no additional geometric state. Square normalization changes representatives, not the underlying square-on-a-segment geometry, and `ac_rfl` only reassociates and commutes a finite scissors sum. For that reason there is no Figure 12.5 for the final substitution: Figures 3 and 4, together with the dependency DAG, already display all spatial information used by the final composition.

8 Lean Formalization

8.1 The public theorem signature

The following is an ASCII transcription of the current compiling source:

```
theorem euclid_proposition_2_12
    [HilbertIncidence Geo]
    [HilbertEuclideanPlane Geo]
    (A B C : Geo.Point)
    (hABC : Not (PrimCollinear Geo A B C))
    (hObtuse : HilbertObtuseAngle Geo B A C) :
    exists D : Geo.Point,
    exists AB0 AB1 AC0 AC1 CB0 CB1 : Geo.Point,
    exists T0 T1 T2 T3 : Geo.Point,
      Geo.Between D A C /\
      HilbertRightAngle Geo A D B /\
      HilbertRightAngle Geo C D B /\
      IsSquare Geo A B AB1 AB0 /\
      IsSquare Geo A C AC0 AC1 /\
      IsSquare Geo C B CB1 CB0 /\
      IsRectangleContainedBy Geo
        T0 T1 T2 T3 D A A C /\
      HilbertScissorsEquicomplementable Geo
        (hilbertParallelogramTerm Geo
          C B CB1 CB0)
        ((hilbertParallelogramTerm Geo
            A B AB1 AB0 +
          hilbertParallelogramTerm Geo
            A C AC0 AC1) +
         (hilbertParallelogramTerm Geo
            T0 T1 T2 T3 +
          hilbertParallelogramTerm Geo
            T0 T1 T2 T3)) := by
```

The theorem is deliberately existential at the representation level. The public interface does not expose:

- the I.12 auxiliary carrier point R;
- the two intermediate squares on DA, DB, DC used by I.47;
- the independent II.4 square representatives;
- the square on DC used to instantiate II.4;
- the auxiliary rectangle representatives inside II.4;
- the upper cut points and diagonal intersections required by the configured rectangle-split APIs.

Only the geometrically meaningful foot D, the three target squares, the target rectangle, and the final content relation escape the wrapper.

8.2 The proposition-local theorem chain

The production module is factored into

```
HilbertObtuseAngle
proposition2_12_obtuse_not_right
```

```
proposition2_12_right_angle_collinear_transport
proposition2_12_perpendicular_foot_exists
proposition2_12_right_angle_swap
proposition2_12_perpendicular_foot_exists_full
proposition2_12_obtuse_perpendicular_foot_between
proposition2_12_pythagorean_package
proposition2_12_same_base_squares_equicomplementable
proposition2_12_normalize_second_pythagoras_DB
proposition2_12_pythagorean_package_common_DB
proposition2_12_ii4_DAC
proposition2_12_scissors_final_bridge
proposition2_12_final_from_normalized_blocks
proposition2_12_normalize_ii4_squares
proposition2_12_final_with_independent_ii4_squares
proposition2_12_parallelogram_L_on_DE
proposition2_12_left_cut_parallelogram
proposition2_12_upper_cut_between
proposition2_12_diagonal_cut_intersection
proposition2_12_parallelogram_cut_exists
proposition2_12_ii4_exists
euclid_proposition_2_12
```

The source order reflects the incremental proof development. The mathematical dependency graph is more naturally read in the construction-first order of the preceding section:

$$\text{obtuse foot} \longrightarrow \text{two I.47 blocks} \longrightarrow \text{II.4 block} \longrightarrow \text{final scissors bridge.}$$

8.3 The public wrapper

The public theorem first calls

```
proposition2_12_pythagorean_package_common_DB
```

and obtains:

$$D - A - C,$$

the two right angles, concrete squares on AB, DA, DB, CB, DC, and the two normalized I.47 relations sharing one DB square.

It then calls

```
proposition2_12_ii4_exists
```

on the same point D. This is an important witness-continuity property: the II.4 block is built on the very perpendicular foot already used by the two I.47 blocks, not on an independently reconstructed collinear configuration.

Finally it calls

```
proposition2_12_final_with_independent_ii4_squares
```

which normalizes the independent II.4 square representatives and invokes the pure final content bridge.

Only the target witnesses are returned.

9 Relationship with Earlier Book II Propositions

Proposition II.12 directly reuses exactly one earlier Book II theorem:

$$\boxed{\text{II.4}}.$$

This is source-faithful. Euclid explicitly invokes II.4 after constructing the perpendicular foot. In the formal proof II.4 supplies the content expansion of the square on DC:

$$S_{DC} \simeq_{\text{ec}} (S_{DA} + S_{AC}) + (R + R).$$

There is no direct dependency on II.11. Proposition II.12 therefore begins a new branch immediately after the construction problem II.11. Its content proof returns to the square-expansion theorem II.4 and combines it with Pythagoras.

There is also no theorem-level shortcut through II.13. The acute-angle analogue is later in Euclid and is not used to prove the obtuse case.

The large configured API of II.4 internally embodies earlier rectangle decomposition machinery, but II.12 does not replace the source proof by calls to II.2 or II.3. Its direct Book II proof dependency remains II.4.

10 Relationship with Book I Infrastructure

The main Book I roles are classical.

- I.12 constructs the perpendicular from B to the carrier AC.
- I.16 is used in the additional formal proof that the foot lies beyond A, giving the exact order $D - A - C$.
- I.47 is applied twice, once to DAB and once to DCB.
- I.46 constructs the concrete squares needed by the internally instantiated II.4 block.

The formal reconstruction also uses the established Hilbert layer:

- strict betweenness and betweenness trichotomy;
- same-ray construction and angle-arm replacement;
- strict angle order, transitivity, transport, and irreflexivity;
- congruence symmetry and transitivity;
- right-angle transport and congruence of all right angles;

- Euclidean parallel uniqueness;
- parallelogram fourth-vertex construction and opposite-side congruences;
- Pasch and parallel crossing order.

These are infrastructure theorems, not new axioms introduced by II.12.

11 Formalization Notes and Hidden Geometric Data

11.1 Obtuse means strict angle order, not numerical measure

The predicate `HilbertObtuseAngle` does not assign a real number to an angle. It gives a synthetic witness: a right angle with the same first arm that is strictly smaller.

Consequently the proof of II.12 never invokes 90°, a cosine, or a trigonometric comparison. The only operations on angles are geometric: congruence, strict order, same-ray transport, opposite extension, and transitivity.

11.2 The order $D - A - C$ is proved, not assumed

This is the source-critical hidden datum of II.12. The classical diagram makes the order visually immediate once the angle at A is obtuse. Lean cannot import that picture as a theorem.

The proof derives $D - A - C$ from:

obtuseness + I.12 perpendicularity + I.16 exterior-angle order + betweenness trichotomy.

Thus the position of the perpendicular foot is part of the mathematical proof object.

11.3 The second right angle needs ray transport

I.12 directly provides one perpendicular direction. For the second Pythagorean triangle the proof needs

$$\angle CDB$$

right. This is not a separate perpendicular construction. Since $D - A - C$, the points A, C lie on the same ray from D, so the right angle is transported from ADB to CDB.

11.4 The II.4 cut witnesses are real geometry

The current II.4 API does not merely accept the assertion that DC is divided at A. It expects explicit upper cut points, diagonal intersections, betweenness data, and parallelograms for the component rectangle splits.

The local parallelogram-cut constructor creates those witnesses by parallel geometry and Pasch. They disappear from the public theorem, but they certify that the formal scissors decomposition corresponds to an actual synthetic geometric dissection.

11.5 Square representatives are noncanonical

Every phrase "the square on XY" hides an existential representative in the formal development. Two separate calls to I.47 may therefore return different squares on DB, and the II.4 construction may return another square on DC than the one already produced by I.47.

The formal proof never assumes uniqueness of square points. It uses `hilbert_square_transport` to prove content equivalence from segment congruence.

11.6 The orientation DA versus AD is not definitional

The first I.47 block naturally produces the square on DA. The II.4 block, under its configured parameter order, produces a square on AD.

These two squares represent the same unoriented classical side, but Lean requires the endpoint reversal to be justified explicitly by segment congruence before square transport can be applied.

11.7 "Greater by" is exact additive content

The final theorem does not introduce a strict order on contents. Instead it proves

$$S_{CB} \simeq_{\mathrm{ec}} S_{AB} + S_{AC} + R + R.$$

This directly witnesses what the excess consists of.

Therefore II.12 does not require:

- an area function;
- positivity of area;
- cancellation;
- an order relation on scissors contents;
- subtraction of figures.

11.8 `ac_rfl` is only bookkeeping

The final bridge uses `ac_rfl` once to expose the subterm

$$S_{DA} + S_{DB}$$

inside a larger formal sum. This operation proves only equality modulo associativity and commutativity of the multiset addition.

It is not a replacement for Euclid's geometric proof. All geometric and content equalities required for the substitution have already been established before this regrouping step.

12 Axiomatic and Semantic Status

The production theorem introduces no new proposition-specific geometric axiom and no new content axiom. It contains no `sorry` or `admit`. The production source compiled cleanly, and the final theorem was subjected to a transitive axiom audit.

The command

```
#print axioms Geometry.euclid_proposition_2_12
```

reported

```
[propext,
 Classical.choice,
 Geometry.bookZero_nullSegment1,
 Quot.sound]
```

These are inherited project dependencies. The precise statement is therefore: II.12 adds no new axiom and no `sorry`; it is not literally axiom-free.

The semantic layers are

$$\begin{array}{c}
\text{incidence/order/congruence and Euclidean parallel geometry} \\
\Downarrow \\
\text{synthetic obtuse-angle order} + \text{I.12 perpendicular construction} \\
\Downarrow \\
\text{proof of the source order } D - A - C \\
\Downarrow \\
\text{two right triangles and two I.47 equicomplementabilities} \\
\Downarrow \\
\text{configured II.4 construction and its scissors content identity} \\
\Downarrow \\
\text{square representative transport} \\
\Downarrow \\
\text{equicomplementability addition, regrouping, and transitivity} \\
\Downarrow \\
\text{II.12 additive equal-content conclusion.}
\end{array}$$

No numerical area function is used. No segment multiplication is used. No trigonometric function or numerical angle measure is used. The modern law-of-cosines formula is an interpretive shadow only.

13 Dependency Summary

The actual production chain is

$$\begin{gathered}
\text{Hilbert incidence, order, congruence, Euclidean parallel layer} \\
\Downarrow \\
\texttt{HilbertObtuseAngle}\text{: right angle strictly below } BAC \\
\Downarrow \\
\text{I.12: perpendicular foot } D \text{ on the carrier } AC \\
\Downarrow \\
\text{I.16 + angle-order transport + trichotomy} \\
\Downarrow \\
D - A - C, \quad \angle ADB \text{ right} \\
\Downarrow \\
\text{same-ray transport: } \angle CDB \text{ right} \\
\Downarrow \\
\text{I.47 on } DAB \quad + \quad \text{I.47 on } DCB \\
\Downarrow \\
\text{square transport: one common } DB \text{ representative} \\
\Downarrow \\
\text{local parallelogram-cut constructor} \\
\Downarrow \\
\text{construct the complete configured II.4 block on } D - A - C \\
\Downarrow \\
\text{II.4: } S_{DC} \simeq_{\mathrm{ec}} (S_{DA} + S_{AC}) + (R + R) \\
\Downarrow \\
\text{normalize } DC \text{ and } DA/AD \text{ square representatives} \\
\Downarrow \\
\text{Common Notion 2 in equicomplementability: add } S_{DB} \\
\Downarrow \\
\text{I.47 substitution + formal regrouping} \\
\Downarrow \\
\texttt{euclid_proposition_2_12}.
\end{gathered}$$

The implementation checkpoint is:

New geometric axiom in II.12:	no
New content axiom in II.12:	no
New proposition-local helper theorem:	yes
New reusable public helper theorem/module:	no
New `HilbertRectangle` theorem:	no
New Interface / Book Zero / Scissors theorem:	no
Directly reuses earlier Book II theorem:	yes, II.4
Conclusion is exact `HilbertScissorsEq`:	no
Conclusion is equal-content / equicomplementability:	yes
Uses exact scissors internally:	yes, through II.4
Uses content cancellation:	no
Uses numerical area:	no
Uses segment multiplication:	no
Uses numerical angle measure / trigonometry:	no
Production source compile:	clean (user verified)
Production theorem axiom audit:	clean baseline
Proposition-local `sorry`/`admit`:	no
Transitive `#print axioms`:	`propext`, `Classical.choice`, `bookZero_nullSegment1`, `Quot.sound`

14 Conclusion

Proposition II.12 preserves Euclid's short top-level proof DAG:

$$\boxed{\text{I.12} \longrightarrow \text{II.4} \longrightarrow \text{add } DB^2 \longrightarrow \text{I.47 twice} \longrightarrow \text{II.12.}}$$

The formal development is longer for two reasons, neither of which changes the mathematical argument.

First, Lean proves the geometric position of the perpendicular foot. The statement

$$D - A - C$$

is derived synthetically from the obtuse-angle hypothesis using I.16, angle order, and betweenness trichotomy. This is the principal hidden geometric fact in the classical diagram.

Second, the existing II.4 interface is deliberately concrete and diagram-explicit. The II.12 module therefore constructs the required parallelogram cuts and normalizes the independently chosen square representatives before performing Euclid's final content substitution.

Once these two formal obligations are discharged, the final scissors calculation is essentially the classical proof. No subtraction, cancellation, numerical area, segment field, coordinate model, cosine, or numerical angle measure is introduced.

The final theorem states the appropriate Book II content relation:

$$\boxed{S_{CB} \simeq_{\text{ec}} (S_{AB} + S_{AC}) + (R + R)}$$

for a rectangle representative R contained by DA, AC. This is the direct synthetic meaning of Euclid's assertion that the square on the side subtending the obtuse angle is greater by twice the indicated rectangle.